cumulus.dat: the 3D cloud file of SMART-G's auxiliary data

Format reference: what the file is made of, what each field holds, and what it is for. File: smartg_auxdata/IPRT/phaseB/grids/cumulus.dat

1. What the file is

cumulus.dat describes **one 3D cloud field**: the output of a large-eddy simulation of cumulus clouds from the I3RC benchmark (Intercomparison of 3D Radiation Codes, phase 2, case 4), quoted at a wavelength of 0.67 µm. It is the cloud the playground renders when no other cloud is chosen. SMART-G reads it with `smartg.atmosphere.read_i3rc_cloud()` and passes it to `Cloud3D`.

It is a plain-text file (425,636 bytes, 20,438 lines, LF line endings). It is **sparse**: the grid has 530,000 cells, but only the 20,435 cells that contain cloud are listed. Every cell that is not listed is clear air.

2. Layout

Line	Content	In this file
1	free-text comment	# I3RC, phase 2; LES Cu (case 4); 0.67 um
2	Nx Ny Nz flag	100 100 53 2
3	dx dy z_0 z_1 ... z_{Nz}	0.0667 0.0667 0.00 0.20 0.40 ... 25.00 30.00 (54 z values)
4 to end	ix iy iz ext reff, one line per cloudy cell	1 2 16 47.891 7.05

```
# I3RC, phase 2; LES Cu (case 4); 0.67 um
100 100 53 2
0.0667 0.0667 0.00 0.20 0.40 0.60 0.80 1.00 1.04 1.08 1.12 1.16 1.20 1.24 1.28 1.32 1.36 1.40 1. ...
1 2 16 47.891 7.05
1 2 17 63.114 8.11
1 2 18 71.673 8.65
...
100 100 23 87.947 9.6
```

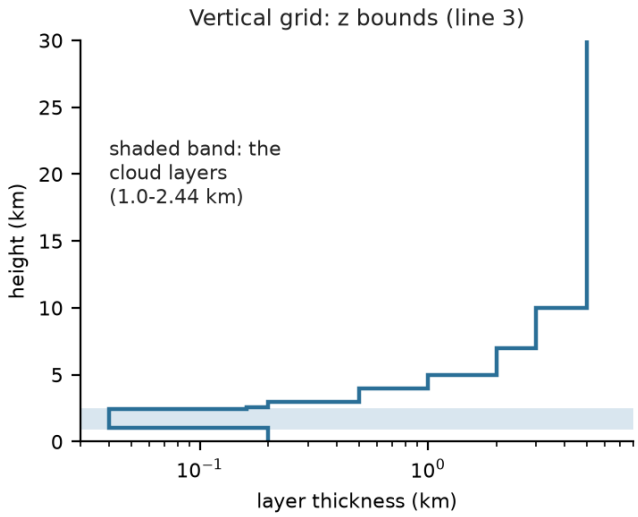
3. The fields and their purpose

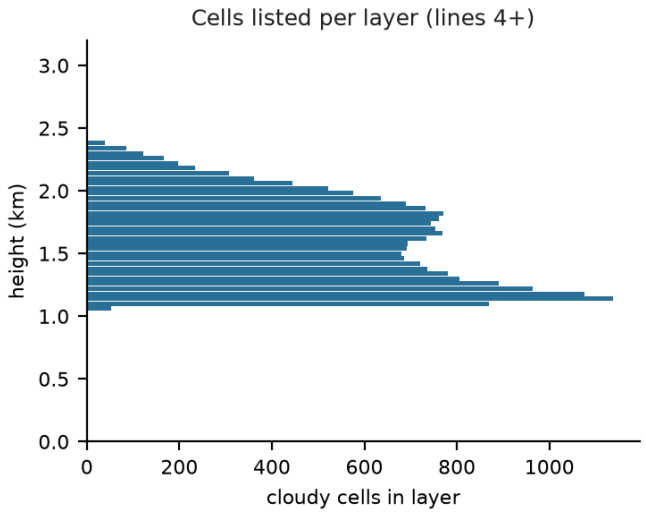
Field	Unit	Meaning	What SMART-G does with it
comment (line 1)	-	Documentation of the origin of the data.	Skipped. The wavelength written in it (0.67 um) is **not** read; SMART-G uses the reference wavelength given in the code (`w_ref` = 670 nm in the playground).
Nx, Ny, Nz (line 2)	cells	Number of cells along x, y and z: 100, 100, 53.	Nx and Ny set the horizontal size of the arrays. Nz is taken from the number of z boundaries on line 3.
flag (line 2)	-	The value 2.	Not read. Probably the number of per-cell properties (ext and reff); this is a guess.
dx, dy (line 3)	km	Horizontal cell size, the same in every cell: 0.0667 km.	Builds x_bounds and y_bounds (Nx+1 and Ny+1 values), centred on 0. The domain is 6.67 km by 6.67 km, from -3.335 to +3.335 km.
z_0 ... z_{Nz} (line 3)	km	Vertical cell boundaries, ascending, Nz+1 values. They need not be equally spaced: here 0.2 km layers up to 1 km, 0.04 km layers through the cloud, then coarser layers up to 30 km.	Builds z_bounds. These are also the layers on which the 1D background atmosphere (molecules, ozone) is evaluated, so the 30 km top is what gives the scene its atmosphere above the cloud.
ix, iy, iz	index	**1-based** integer position of the cell along x, y and z.	Converted to 0-based array indices; the cell goes to element [iz, iy, ix] of the arrays.

Field	Unit	Meaning	What SMART-G does with it
ext	km^{-1}	Extinction coefficient of the cell at the reference wavelength: the chance per kilometre of path that light is scattered or absorbed. It measures how much cloud the cell holds. Always positive in the file.	Stored in the array *ext*. A cell is cloud only where ext > 0. SMART-G scales it to other wavelengths with the table wc_sol.nc (about 1% between 440 and 670 nm).
reff	µm	Effective radius of the cloud droplets in the cell (area-weighted mean radius). Always positive in the file.	Stored in the array *reff*. It selects the droplet optics, the single-scattering albedo and the phase function, from wc_sol.nc by interpolation. Values outside 5-30 µm are not accepted, so the playground raises smaller ones to 5 µm (the file's minimum is 4.0).

4. What this particular file holds

Property	Value
Grid (z, y, x)	53 x 100 x 100 = 530,000 cells
Horizontal cell size, domain	0.0667 km; 6.67 km x 6.67 km
Vertical extent	0 to 30 km; 0.2 km layers below 1 km, 0.04 km layers from 1.0 to 2.44 km, then 0.16 to 5 km layers up to 30 km
Cloudy cells listed	20,435 (3.9% of the grid), in layers iz = 6 to 41, i.e. 1.0 to 2.44 km
ext (cloudy cells)	0.041 to 196.4 km^{-1}, median 68.4
reff (cloudy cells)	4.0 to 14.4 µm, median 8.5
Largest vertical optical depth	about 151
Order of the cell lines	ix slowest, then iy, iz fastest; sorted, no duplicates
Number format	ext up to 3 decimals, reff up to 2, trailing zeros dropped (for example 7.05, 8.7)





5. How SMART-G turns the file into its input

`read_i3rc_cloud()` reads lines 2 and 3 for the grid, then allocates two arrays of shape (Nz, Ny, Nx) filled with zeros and writes *ext* and *reff* into the listed cells. It returns a dataset with *ext*, *reff* and the coordinates x_bounds, y_bounds and z_bounds. Then:

Step	Object	Uses
1	Cloud3D("wc", w_ref=670, ds=...)	ext and reff of the cells with ext > 0; droplet optics from wc_sol.nc
2	Grid3D(x_bounds, y_bounds, z_bounds, ...)	the three bounds arrays, plus a periodic or empty-margin horizontal boundary
3	Atm3D(atm_1d, grid_3d, comp_3d=[cloud])	mixes the cloud with the background atmosphere in the cloudy cells

6. What the file does not contain

• **Position on Earth.** Only the cell sizes are stored; the horizontal grid is centred on 0 by the reader.
• **Wavelength.** The 0.67 µm is only in the comment; the code decides it.
• **Phase function and albedo.** They come from the table wc_sol.nc, by effective radius.
• **Water content and droplet number.** Only the two derived optical quantities, ext and reff, are stored.
• **Time.** The file is one snapshot.

Compiled from the raw file and from the source of SMART-G 2.0.0b1 (read_i3rc_cloud, Cloud3D, Grid3D). The meaning of the flag on line 2 is not confirmed.